Train a Regression Model

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

Teach line fitting with no missing validation targets.

Supply the observations implied by W2/W3,interpret coefficient units and distinguish candidate search from global optimization or biological law.

Keep one habit: Require training-only selection,unchanged state and same-case baseline evidence.

MAIN LESSON · SECTION 1

Prepare the connected experiment

> Keep prior exposure, shared practice and independent evidence distinct.

Goal and main teaching plan

Primary target: Fit a line from a declared **training candidate set**

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Materials and prerequisite

Teach two 45-minute sessions. This week focuses on **fitting** a line. Model the worked example first, collect the new W 4 case independently before feedback, and use the fresh retry to check transfer.

Award credit for traceable reasoning and bounded claims, including justified uncertainty.

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Square each residual, sum, then divide by the number of evaluated examples.

Printed student lesson and workbook

Paper, pencils and a calculator for fractions

The fictional examples and a blank **evidence** table

Optional Python 3 editor; no external packages or AI accounts required

Opening retrieval

Without opening the old lesson, explain one key idea from Absolute Error and MAE. Give an example and a mistake that the idea helps you avoid.

Look for: Calculate MAE MSE and RMSE

Reference example, not the only acceptable wording: Teacher explanation: (1+2+3)/3=2 cm.

Without opening the old lesson, explain one key idea from Regression and Residuals. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish numeric **prediction** from classification

Reference example, not the only acceptable wording: Teacher explanation: Residuals -1, 1, 0 cm; the first record is overpredicted.

Without opening the old lesson, explain one key idea from Accuracy and Imbalance. Give an example and a mistake that the idea helps you avoid.

Look for: Explain why overall accuracy can hide failure on a less common class

Reference example, not the only acceptable wording: The **training**-majority rule always predicts clear. Matrix [[16,0],[4,0]] gives 16/20=80 percent, FP=0,FN=4.

It misses all four blocked cases, so the overall score hides complete failure on that evaluated class.

End goal

By the end, I can explain train a regression model using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Absolute Error and MAE

Regression and Residuals

Accuracy and Imbalance

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Training points (x,y)=(1,3),(2,5),(3,7). Compare y=x+1 and y=2x+1 by **training** MAE; select one, then predict held-out x=4, actual=10. Explain what was fitted.

Concept to explain

A fitted relationship should be interpreted, tested on unseen values, and compared with a simple baseline.

Term	Meaning
coefficient	learned relationship weight
intercept	prediction when inputs are zero
slope	predicted change per input unit
generalization	performance on unseen cases

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach a limited candidate search with all validation observations visible

> Keep summed training error,held-out averages and coefficient units distinct.

Main fitting contract and complete source roles

A line predicts y as m*x+b. In this classroom search we compare a small declared set of **slope**s and **intercept**s; this is not a full least-squares solver over all possible lines.

Use **training** error to choose a candidate, then freeze it before validation.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason. Earlier weeks supplied the model-building **process**; this week examines **fitting** a line.

Role / ID	x in days	Observed height y in cm	Use
Training T1	0	1	Fit among declared A/B/C candidates

Role / ID	x in days	Observed height y in cm	Use
Training T2	1	3	Fit among declared A/B/C candidates
Training T3	2	5	Fit among declared A/B/C candidates
Validation V1	3	6	Score the frozen line and baseline
Validation V2	4	10	Score the frozen line and baseline

Training pairs are (0, 1),(1, 3),(2, 5). Candidate A has m 1 b 1, B m 2 b 0, C m 2 b 1. **Prediction**s are A [1, 2, 3], B [0, 2, 4], C [1, 3, 5]. **Training** squared-error sums are 5, 3, 0 respectively, so choose C.

Ties would use the earlier listed candidate.

Trace the calculation on paper before running this optional Python check. Use the stated fictional **input**s and compare intermediate values. The code checks this example; your independent workbook case uses different **evidence**.

Term	Meaning
Slope	Change in predicted target per unit change in x
Intercept	Predicted target when x is zero
Candidate set	The declared model choices searched
Fitting	Selecting model values using training evidence

Trace validation and baseline evidence

Fitting a line at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Line A	Slope 1 intercept 1	Training SSE 5
Line B	Slope 2 intercept 0	Training SSE 3
Line C	Slope 2 intercept 1	Training SSE 0

For separate validation pairs (3, 6),(4, 10), C predicts [7, 9], giving residuals [-1, 1], MAE 1 and MSE 1. The training-mean constant baseline is 3 and gives residuals [3, 7], MAE 5 and MSE 29. The fitted line wins on this small evaluation, without proving performance elsewhere.

Predict at x 3 and x 4 and calculate MAE/MSE. Explain the steps or evidence behind your response.

Main training [(0,1),(1,3),(2,5)] chooses (m,b)=(2,1); main validation [(3,6),(4,10)] gives predictions [7,9], residuals [-1,+1] cm, MAE1cm and MSE1cm². Training mean baseline3cm gives residuals [3,7], MAE5cm,MSE29cm². Picture uses only day3 observation6, later changed to8: prediction7 stays frozen, absolute error1cm in both cases, while residual changes-1→+1cm.

Main validation is a tiny extrapolation beyond training days0–2; it does not justify a biological growth law or long-range forecasts.

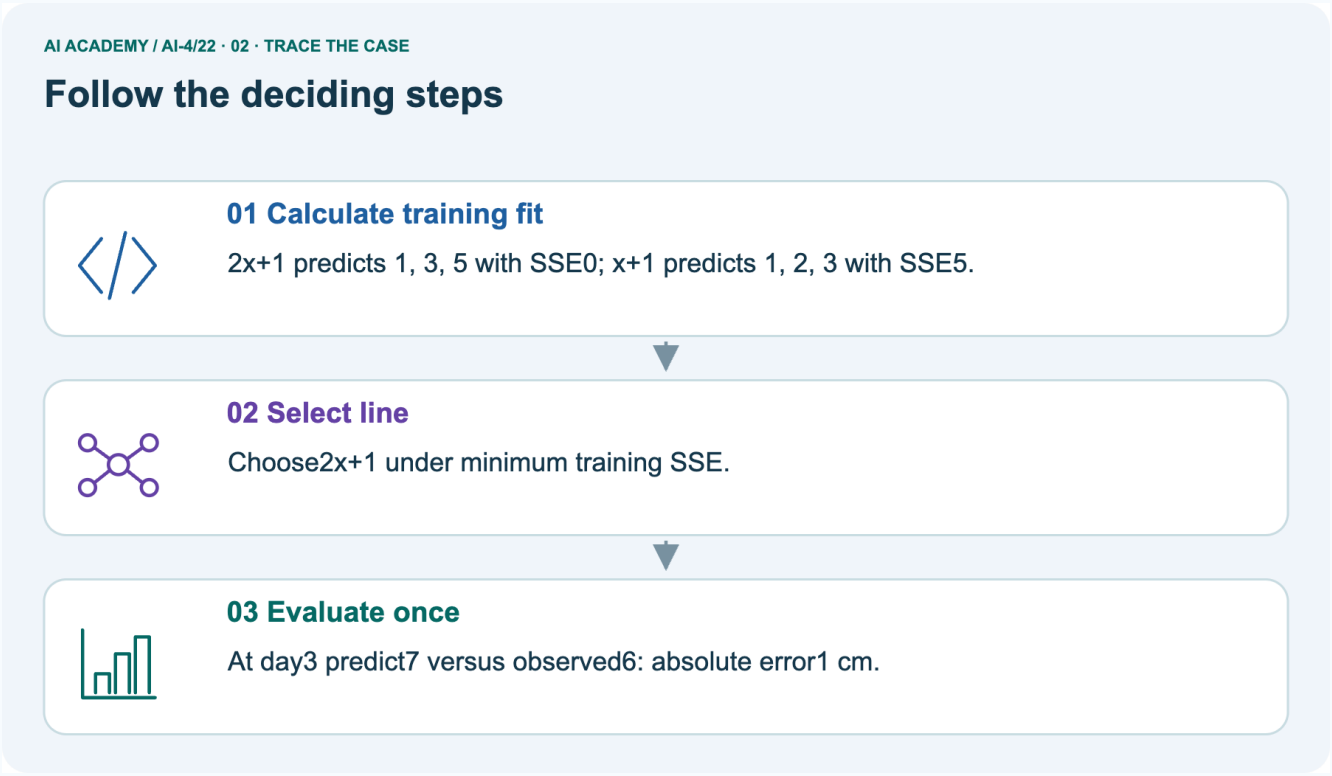
Read separate pictured day3observation

Fit a toy growth line from **training** pairs, then leave evaluation labels out of **fitting**.

Training(day, height):(0, 1),(1, 3),(2, 5). Candidates y=2x+1 and y=x+1. Evaluation pair:(3, 6).

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Trace unchanged prediction under changed observation



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Calculate training fit	2x+1 predicts 1, 3, 5 with SSE0; x+1 predicts 1, 2, 3 with SSE5.
2. Select line	Choose2x+1 under minimum training SSE.
3. Evaluate once	At day3 predict7 versus observed6: absolute error1 cm.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Read the picture in order. At day3 predict7 versus observed6: absolute error1 cm. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

An exact fit to three toy pairs does not establish a biological growth law or justify long-range extrapolation.

Change: Evaluation **observation** becomes8 instead of6.

Prediction: Absolute evaluation error stays1 cm, but residual changes from-1 to+1.

Reason: **Prediction**7 remains frozen; changed **observation**8 reverses the error direction.

▼ Additional supplied-rule inference example

Supplied fitted rule: height=2 times age+1. Held-out **observation**s are (3,8),(4,9).

Predict and evaluate; distinguish inference.

Predictions 7,9; errors 1,0; MAE=.5 cm. Applying a supplied equation does not show how it was fitted.

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Trace the information, fitted behavior, and evaluation proof separately.

QUESTION Run a starter regression and explain **slope**, **prediction**, residual, and MAE.

MODEL IDEA A fitted relationship should be interpreted, tested on unseen values, and compared with a simple baseline.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Model predicts height = 4 + 2 × weeks. **Slope** 2 means two predicted centimeters per additional week in this data range.

Draw a fresh data → model → **prediction** → evaluation trace:

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

STEP 1 Model predicts height = 4 + 2 × weeks.

STEP 2 **Slope** 2 means two predicted centimeters per additional week in this data range.

STEP 3 At week 5, **prediction** is 14 cm; if actual is 16, residual is +2 cm.

STEP 4 Evaluate unseen plants and compare test MAE with the median baseline.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

Concrete teaching case

Exact explanation to model

Explain the added worked example

Key terms

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

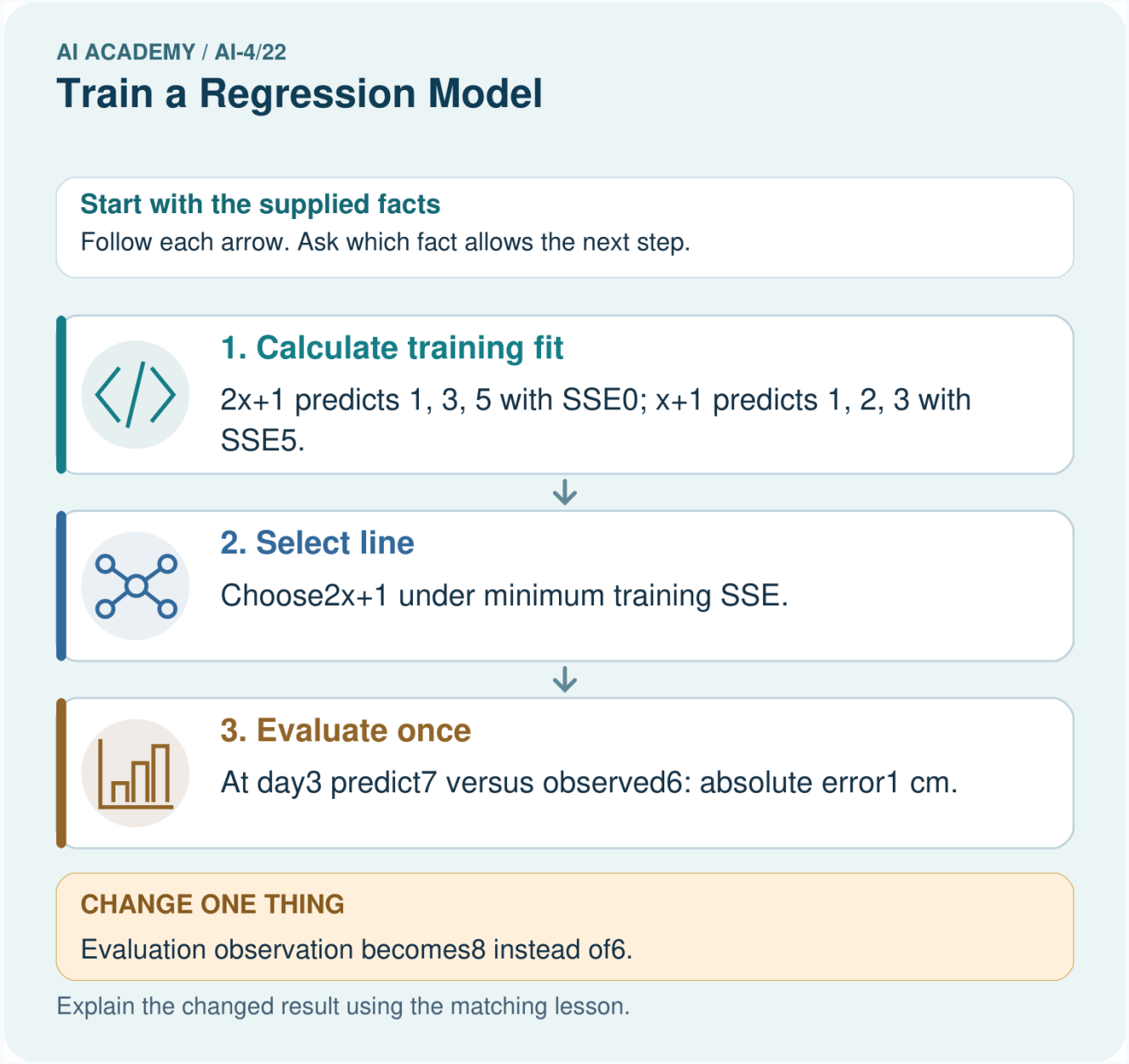
These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Train a Regression Model

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Guide the calculation, then remove help

> Collect each learner's prediction before showing its key.

Current completion task and checked key

Completion task: Predict at x 3 and x 4 and calculate MAE/MSE.

Teacher check: Teacher explanation: **Prediction**s 7, 9; residuals -1, 1; MAE 1 MSE 1.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for **Fitting** a line. Proposed explanation: "A final answer is enough". Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Ask for the relevant counts, units, procedure or saved record. For this week: A 5 B 3 C 0; choose m 2 b 1.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial question and changed-case reasoning

May day3's evaluation label choose the candidate line? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

No under this **fitting** plan. Keep evaluation **evidence** separate and report the frozen line's1 cm error. Changed-case check: Absolute evaluation error stays1 cm, but residual changes from-1 to+1.

Prediction7 remains frozen; changed **observation**8 reverses the error direction.

Changed-case reasoning: Absolute evaluation error stays1 cm, but residual changes from-1 to+1. **Prediction**7 remains frozen; changed **observation**8 reverses the error direction.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep each learner explaining

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Earlier wording, alternatives and picture transcripts

Compact alternative: this two-session sequence covers selected core tasks. Use the three-session weekly plan for the full lesson, independent transfer, feedback and project work; allow extra time for extensions.

Demonstrate and guide practice

Board sequence

Write the given facts: Supplied fitted rule: height=2 times age+1. Held-out **observation**s are (3,8),(4,9).

Write the question to explain: Predict and evaluate; distinguish inference.

Reveal after **prediction**s: **Prediction**s 7,9; errors 1,0; MAE=.5 cm. Applying a supplied equation does not show how it was fitted.

Use the earlier transfer question as additional practice. Reserve the selected fresh m3/b2line fitting and validation task for the first individual attempt and record support used.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

Guided response: **Prediction**s 7, 9; residuals -1, 1; MAE 1 MSE 1. Exit response: **Training** covers a limited range; the relationship and conditions may change outside it.

Case-specific teaching sequence

Stage · Teacher action

> **Stage:** Session 1 start to 7 minutes
>
> **Teacher action:** Use the prerequisite question: How is MSE calculated? Check the explanation before proceeding.

> **Stage:** 7 to 18 minutes

Teacher action: Read the opening and model the worked evidence. Ask students to identify slope in context.

Stage: 18 to 32 minutes

Teacher action: Complete W 1 and W 2 with support: trace candidates; score validation. Require the evidence, not only the final response.

Stage: 32 to 45 minutes

Teacher action: Read the visual row by row. Collect the guided explanation and compare reasoning in pairs. Trace the Python check after the paper calculation.

Stage: Session 2 start to 8 minutes

Teacher action: Retrieve the distinction between slope and intercept. Invite a counterexample to an overconfident claim.

Stage: 8 to 20 minutes

Teacher action: Discuss the application and complete W 3: Fit the training constant and score it on validation.

Stage: 20 to 35 minutes

Teacher action: Collect W4 independently before giving hints. Assess these outcomes: Fit a line from a declared training candidate set; Separate training loss from validation comparison. Record any support used.

Stage: 35 to 45 minutes

Teacher action: Use W 5 and the exit response to assess limits. Give targeted feedback, then the fresh retry. Save evidence of independent understanding.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Keep predictions, observed or supplied outcomes and unmeasured cases separate.

Supplied inputs and checked guidance

Complete main candidate SSE5/3/0selects(2,1);validation6/10gives predictions7/9,MAE1,MSE1;training-mean baseline3givesMAE5,MSE29. Tied one-point fixture chooses first listed without proving its slope. Empty data unavailable;far100predicts201algebraically without validated biological meaning. Independent SSE5,3,0selects(m3,b2);validation predictions11/14,residuals-1/+1,MAE1. No validation label enters fitting.

Predict, investigate and record

1. 1. Write the hypothesis and expected **evidence** before running.
2. 2. Regression Run: trace code, interpret **coefficient**s with units, predict unseen values, and compare MAE with baseline.
3. 3. Run one normal, one boundary, one missing, and one shifted case.
4. 4. Calculate the metric from raw counts or row-level errors.
5. 5. Interpret what changed, what did not, and what remains unsafe or unknown.

Keep the experiment ledger

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose the current misconceptions

A final answer is enough: Ask for the relevant counts, units, procedure or saved record. For this week: A 5 B 3 C 0; choose m 2 b 1.

A stronger claim makes a stronger project: **Training** covers a limited range; the relationship and conditions may change outside it.

The new case can be copied from the example: The independent task changes the **evidence**. Require a fresh explanation: SSE 5, 3, 0 selects m 3 b 2. **Prediction**s 11, 14; residuals -1, 1; MAE 1.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows	Next teaching action
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Keep the test conditions fixed, record every result, and change only one planned factor.

Make the lab runnable

Use the supplied complete investigation for shared practice. Reserve the selected fresh m3/b2line fitting and validation task for the individual first attempt before hints or feedback; earlier transfer questions remain additional practice.

Diagnose and reteach

MISCONCEPTION TO REPAIR A positive **slope** proves one variable causes the other.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the selected experiment transfer

> Keep this key until after a first attempt; record any help and prior exposure.

Selected independent task and checked key

Selected case: **Training** (0, 2),(1, 5),(2, 8). Candidates m 2 b 2, m 3 b 1, m 3 b 2. Select by SSE, then score (3, 10),(4, 15) with MAE.

Teacher explanation: SSE 5, 3, 0 selects m 3 b 2. **Prediction**s 11, 14; residuals -1, 1; MAE 1.

Judge trace, scope and support

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Trace candidates

Give the three **training** SSE values and selected line.

Teacher explanation: A 5 B 3 C 0; choose m 2 b 1.

W2 Score validation

Predict at x 3 and x 4 and calculate MAE/MSE.

Teacher explanation: **Prediction**s 7, 9; residuals -1, 1; MAE 1 MSE 1.

W3 Compare the baseline

Fit the **training** constant and score it on validation.

Teacher explanation: **Training** mean 3; residuals 3, 7; MAE 5 MSE 29.

W5 Limit extrapolation

Why is an exact **training** line not proof of correct **prediction**s at x 100?

Teacher explanation: **Training** covers a limited range; the relationship and conditions may change outside it.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

Predictions 7,9; errors 1,0; MAE=.5 cm. Applying a supplied equation does not show how it was fitted.

Additional transfer practice question

A regression model has validation MAE 2.4 cm; a **training**-mean baseline has validation MAE 2.0 cm on the same examples. Which performs better on this metric?

Reasoned practice answer

The baseline is better because lower MAE is preferable: 2.0 < 2.4 cm. Investigate the model before claiming improvement.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “A trained model necessarily outperforms a simple baseline.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

First model errors are 1,2,3, MAE=2; second errors are 0,0,0, MAE=0. Select 2x+1. Test **prediction**=9 and absolute error=1.

This selects among two supplied candidates; it is not a derivation of unrestricted linear-regression **fitting**.

Explanations for the additional workbook examples

W1 Trace candidates

W2 Score validation

W3 Compare the baseline

W4 Apply independently

Training (0, 2),(1, 5),(2, 8). Candidates m 2 b 2, m 3 b 1, m 3 b 2. Select by SSE, then score (3, 10),(4, 15) with MAE.

Teacher explanation: SSE 5, 3, 0 selects m 3 b 2. **Prediction**s 11, 14; residuals -1, 1; MAE 1.

W5 Limit extrapolation

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and revisit the experiment

> Retain failures and assess a fresh parallel case after help.

Project evidence

Add the regression experiment to the project appendix only as a separate, clearly limited model.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Support and selected reassessment

Use a tiny labelled dataset, explicit units and a named positive class. Stretch: require a baseline, held-out evaluation and subgroup or shifted-condition analysis with denominators.

Provide a blank **evidence** table and allow a verbal explanation before writing. Highlight the given conditions without supplying the W 4 conclusion. Use **slope** and **intercept** in full sentences.

Extension: alter one condition in the worked example and explain which conclusion changes and which remains supported.

For independent W 4, require: SSE 5, 3, 0 selects m 3 b 2. **Prediction**s 11, 14; residuals -1, 1; MAE 1. Fresh retry **training** (0, 0),(1, 2),(2, 4), candidates m 1 b 0 and m 2 b 0: SSE 5 and 0.

At validation (3, 7), chosen **prediction** 6 gives residual 1. Mark each criterion as independent, supported or not yet demonstrated. Reassess the reasoning after feedback; do not replace the first attempt silently.

Delayed return

Suggested return points in this level: ai-4/23, ai-4/25, ai-4/28. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	

Record	Teacher evidence
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge and reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 23: Hold Out Test Data. Retrieve today's **evidence** and identify what carries forward and what is new.

Next is Week 23: Hold Out Test Data. Carry forward the exact training-selected line, baseline, candidate order, source assignments and validation predictions. Keep later development choices separate from a protected final check.

▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

Responsible Plant Classifier

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Extension · Teach the least-squares formula lab**

REFERENCE · SECTION 7

Extension · Teach the least-squares formula lab

> Keep this separate weeks-based formula method and its visible one-case check distinct from the main candidate-search activity.

Complete extension and checked code-change key

Fit and test a straight-line regression model

Estimate **slope** and **intercept** from **training** pairs using least squares. Ages are weeks and heights are cm. Test targets are excluded from **fitting**; **training** ages must include distinct values.

```
"""Fit a line to training pairs and evaluate a separate example."""
train = [(1, 3), (2, 5), (3, 7)]
if not train:
    raise ValueError("Training examples are required")
n = len(train)
x_mean = sum(x for x, y in train) / n
y_mean = sum(y for x, y in train) / n
num = sum((x - x_mean) * (y - y_mean) for x, y in train)
den = sum((x - x_mean) ** 2 for x, y in train)
if den == 0:
    raise ValueError("Distinct training ages are required")
slope = num / den
intercept = y_mean - slope * x_mean
test = [(4, 10)]
if not test:
```

```
    raise ValueError("Test examples are required for MAE")
preds = [slope * x + intercept for x, y in test]
mae = sum(abs(y - pred) for pred, (x, y) in zip(preds, test)) / len(test)
print(slope, intercept, preds, mae)
```

Expected **output**: 2.0 1.0 [9.0] 1.0

The fitted line is height = 2 times age + 1. At four weeks it predicts 9 cm; the observed height of 10 cm gives an absolute error and one-case MAE of 1 cm. **Fitting** a line and applying a supplied line are different activities.

One held-out case is very limited **evidence**.

Independent code change

Change the held-out height from 10 to 8 without changing the **training** data. Predict the fit, **prediction** and MAE.

Teacher answer: **Slope** 2, **intercept** 1, **prediction** 9 cm and MAE 1 cm are unchanged. The residual sign changes under the actual-minus-predicted convention.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 8

Reference · Alternative planning checklist

> Use alongside the main route without repeating the whole cycle.

Planning and assessment reference

Teaching target: Fit a line from a declared **training candidate set**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

Before class, solve the supported practice: Predict at x 3 and x 4 and calculate MAE/MSE.

Reasoned practice answer: Teacher explanation: **Prediction**s 7, 9; residuals -1, 1; MAE 1 MSE 1.

Check every learner before discussion: Use the worked case for **Fitting** a line. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Ask for the relevant counts, units, procedure or saved record. For this week: A 5 B 3 C 0; choose m 2 b 1.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Evaluation **observation** becomes8 instead of6. Require a **prediction** and a reason before comparing. Expected result: Absolute evaluation error stays1 cm, but residual changes from-1 to+1.

Reason: **Prediction**7 remains frozen; changed **observation**8 reverses the error direction.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback and evidence records**

REFERENCE · SECTION 9

Reference · Feedback and evidence records

> Keep supported rehearsal separate from independent performance.

Record and repair

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

▼ Earlier wording, alternatives and picture transcripts

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

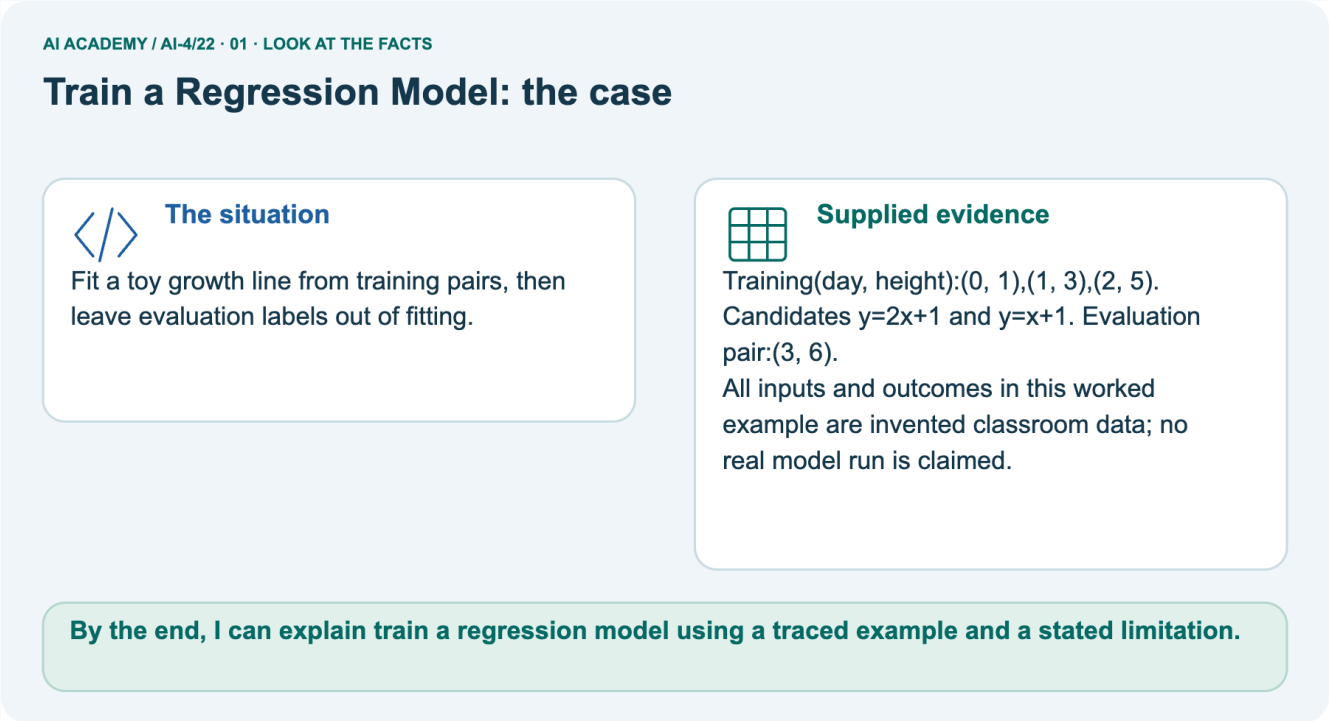
▼ Reference · Teach with the six pictures

REFERENCE · SECTION 10

Reference · Teach with the six pictures

> Choose one picture for a specific reasoning move.

01. Train a Regression Model: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

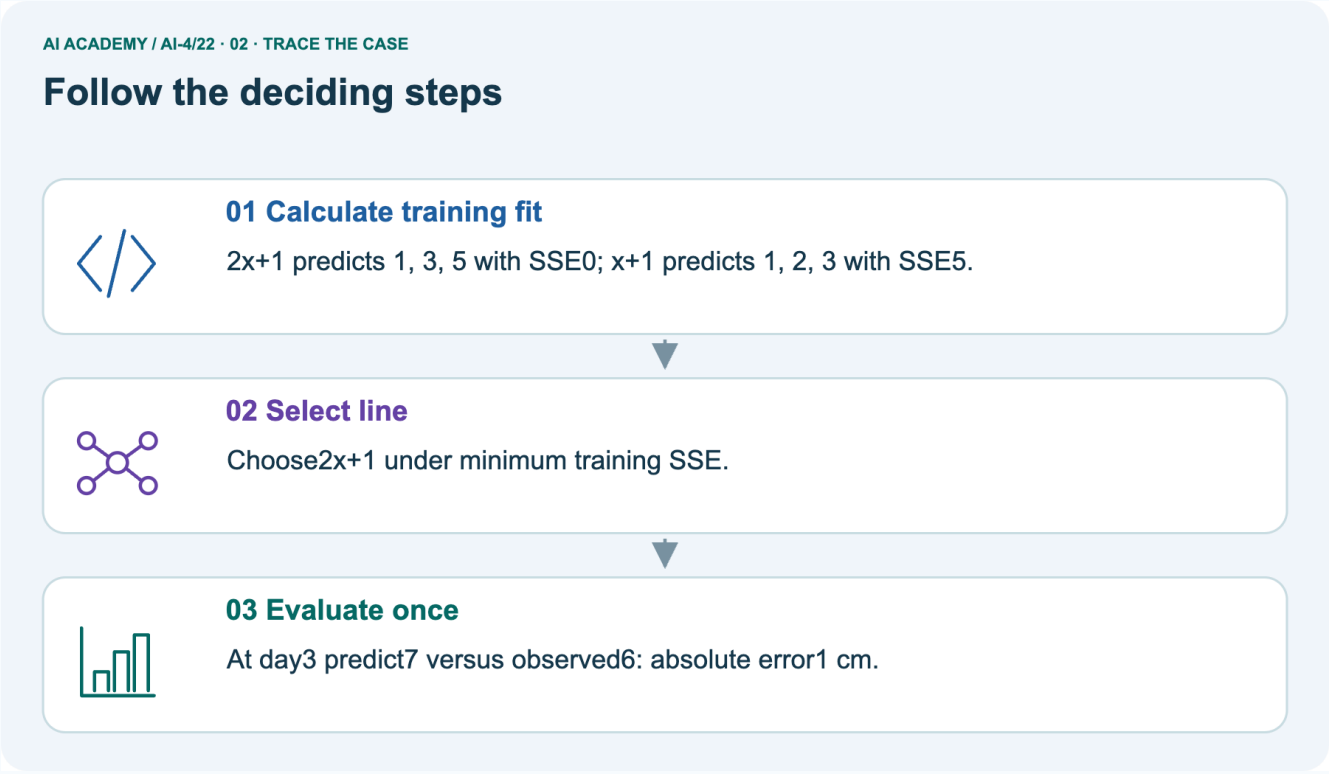
Fit a toy growth line from **training** pairs, then leave evaluation labels out of **fitting**. Supplied facts: **Training**(day, height):(0, 1),(1, 3),(2, 5). Candidates y=2x+1 and y=x+1. Evaluation pair: (3, 6).

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



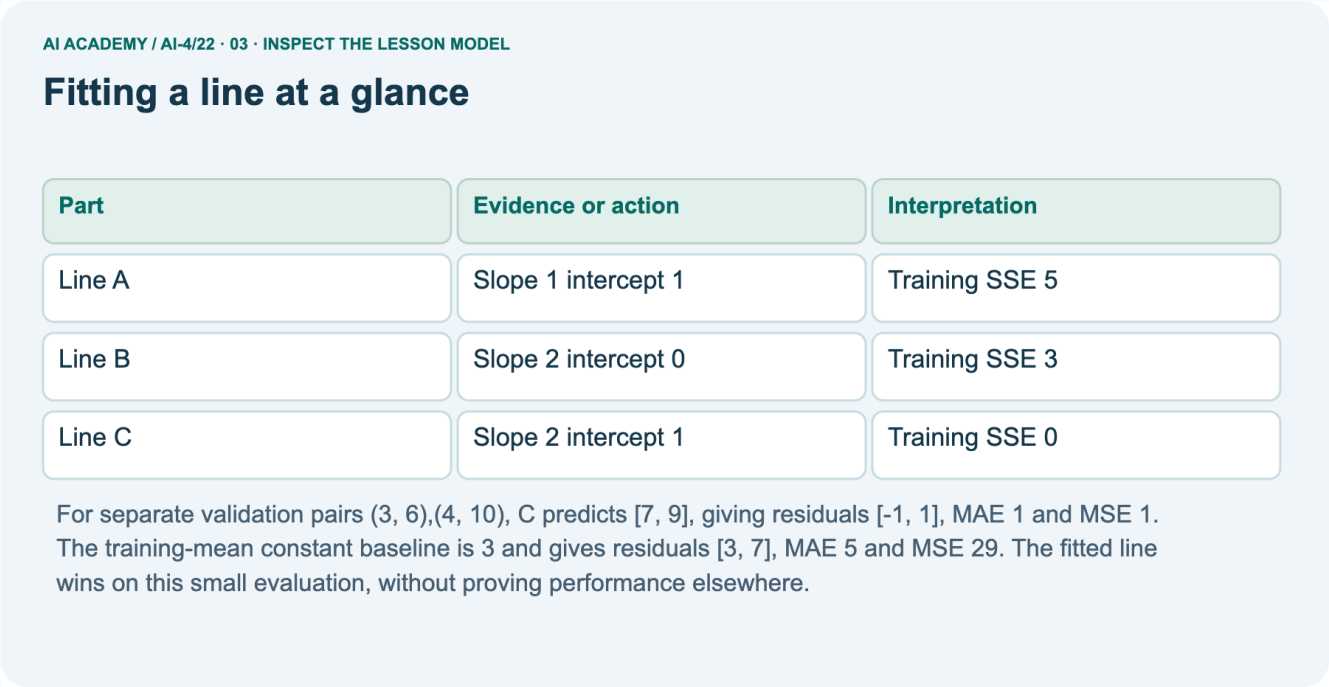
Follow the labelled picture with the matching explanation. Open the vector diagram

Calculate **training** fit: 2x+1 predicts 1, 3, 5 with SSE0; x+1 predicts 1, 2, 3 with SSE5. → Select line: Choose2x+1 under minimum **training** SSE. → Evaluate once: At day3 predict7 versus observed6: absolute error1 cm.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Fitting a line at a glance



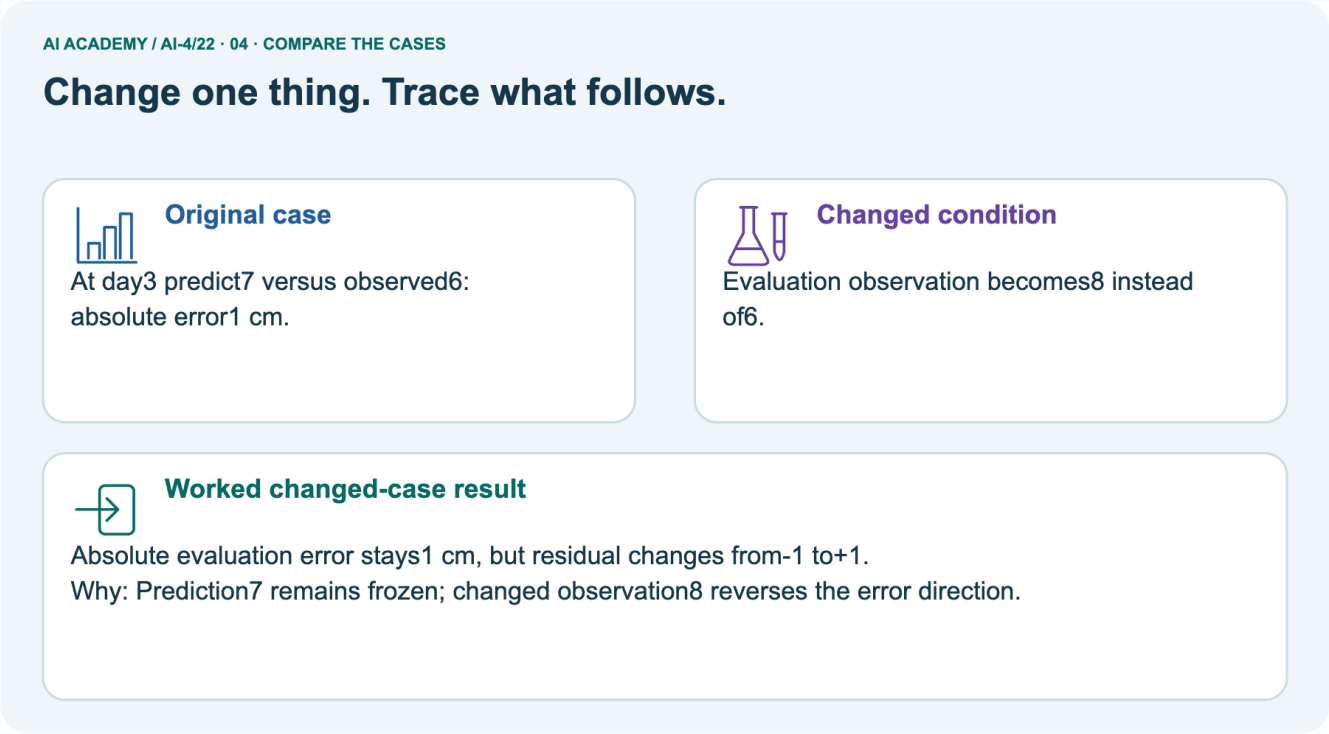
Follow the labelled picture with the matching explanation. Open the vector diagram

Fitting a line at a glance. Line A | **Slope** 1 **intercept** 1 | **Training** SSE 5. Line B | **Slope** 2 **intercept** 0 | **Training** SSE 3. Line C | **Slope** 2 **intercept** 1 | **Training** SSE 0

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: At day3 predict7 versus observed6: absolute error1 cm. Changed condition: Evaluation **observation** becomes8 instead of6. Expected result: Absolute evaluation error stays1 cm, but residual changes from-1 to+1.

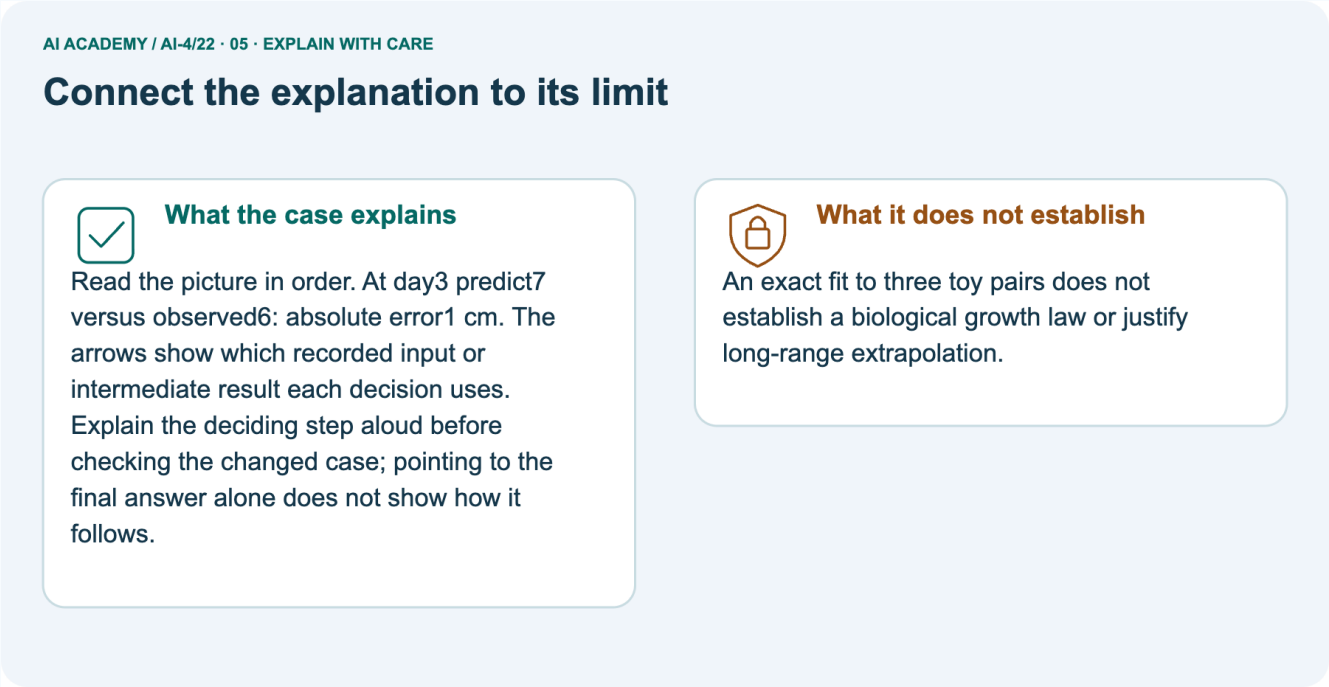
Reason: **Prediction**7 remains frozen; changed **observation**8 reverses the error direction.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: **Prediction**7 remains frozen; changed **observation**8 reverses the error direction.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. At day3 predict7 versus observed6: absolute error1 cm. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

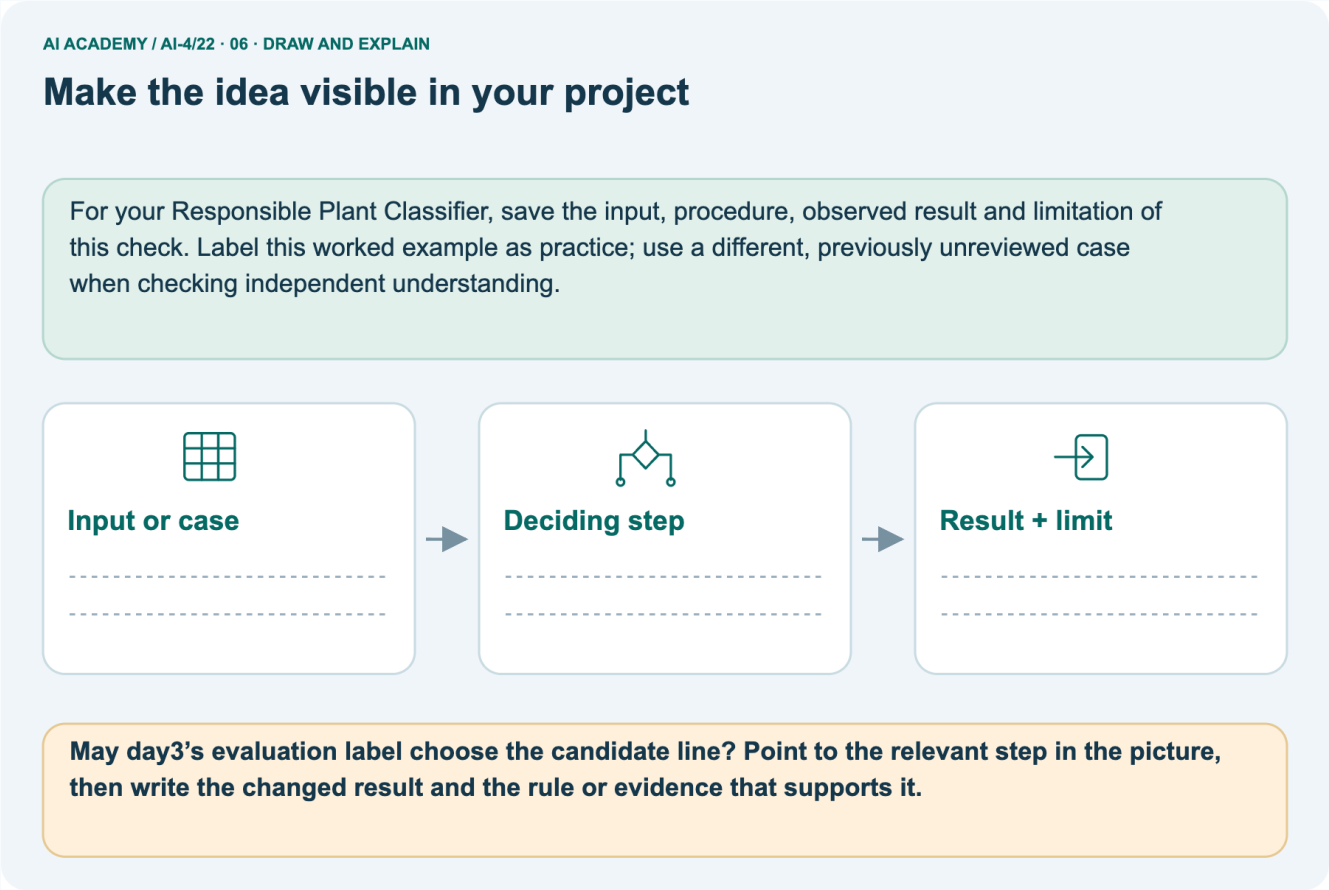
Limit: An exact fit to three toy pairs does not establish a biological growth law or justify long-range extrapolation.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: An exact fit to three toy pairs does not establish a biological growth law or justify long-range extrapolation.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: May day3's evaluation label choose the candidate line? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: No under this **fitting** plan. Keep evaluation **evidence** separate and report the frozen line’s1 cm error. Changed-case check: Absolute evaluation error stays1 cm, but residual changes from-1 to+1.

Prediction7 remains frozen; changed **observation**8 reverses the error direction.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Train a Regression Model: the case

02. Follow the deciding steps

03. Fitting a line at a glance

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.